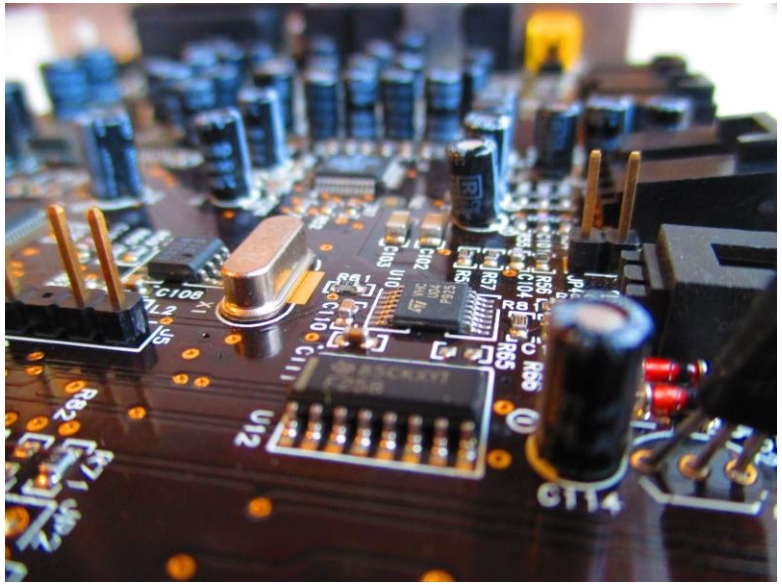
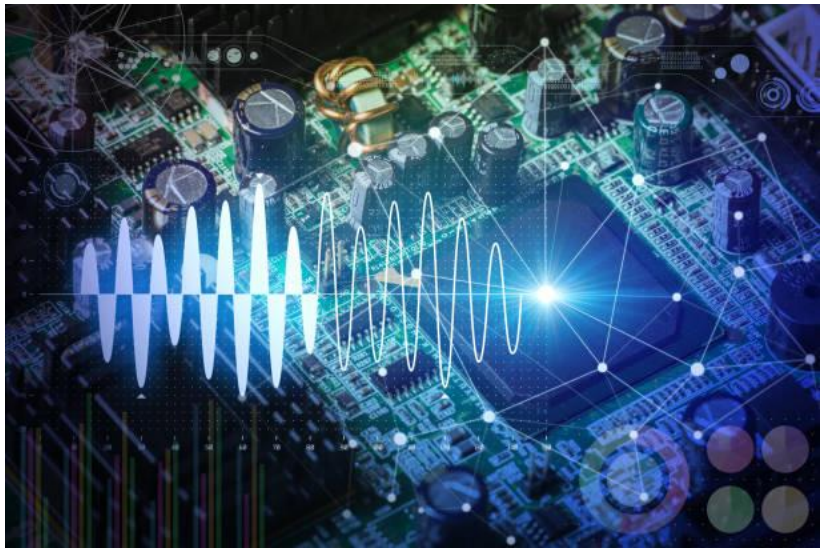
EE3301 Analog Electronics





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DC POWER SUPPLIES AND REGULATION CIRCUITS

1. Diodes
2. Half Wave Rectification and Full wave Rectification
3. Clamping and Clipping
4. Filtering and Regulation

Diodes

Types of materials

1. Conductor

is defined as materials that allow electricity to flow through them easily.

Eg:- Copper, Gold, Iron, Aluminum



2. Insulator

is a material that does not conduct electrical current

Eg:- Plastic, Wood, Glass



Dry wood

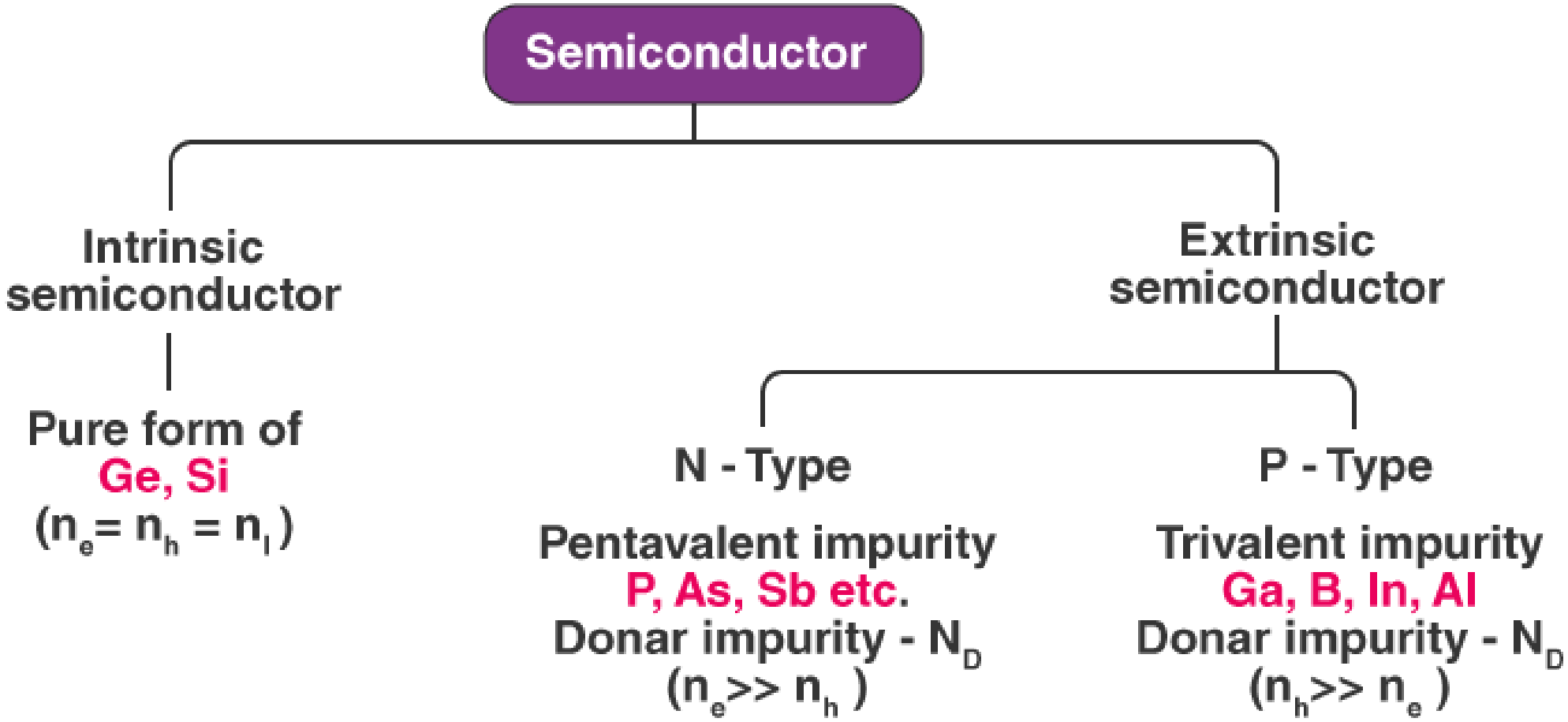


Glass

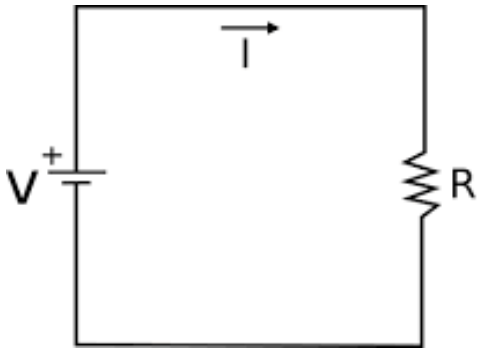


Rubber

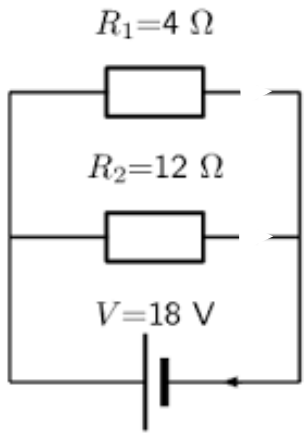
3. Semiconductor


Semiconductor
Intrinsic semiconductor
Pure form of
Ge, Si
($n_e = n_h = n_i$)
Extrinsic semiconductor
N - Type
Pentavalent impurity
P, As, Sb etc.
Donar impurity - N_D
($n_e >> n_h$)
P - Type
Trivalent impurity
Ga, B, In, Al
Donar impurity - N_D
($n_h >> n_e$)

Q1) When V= 8.8 V and R= 2.2 KΩ, Calculate the current through the resistor.



Q2) Calculate the total current drawn from the battery.

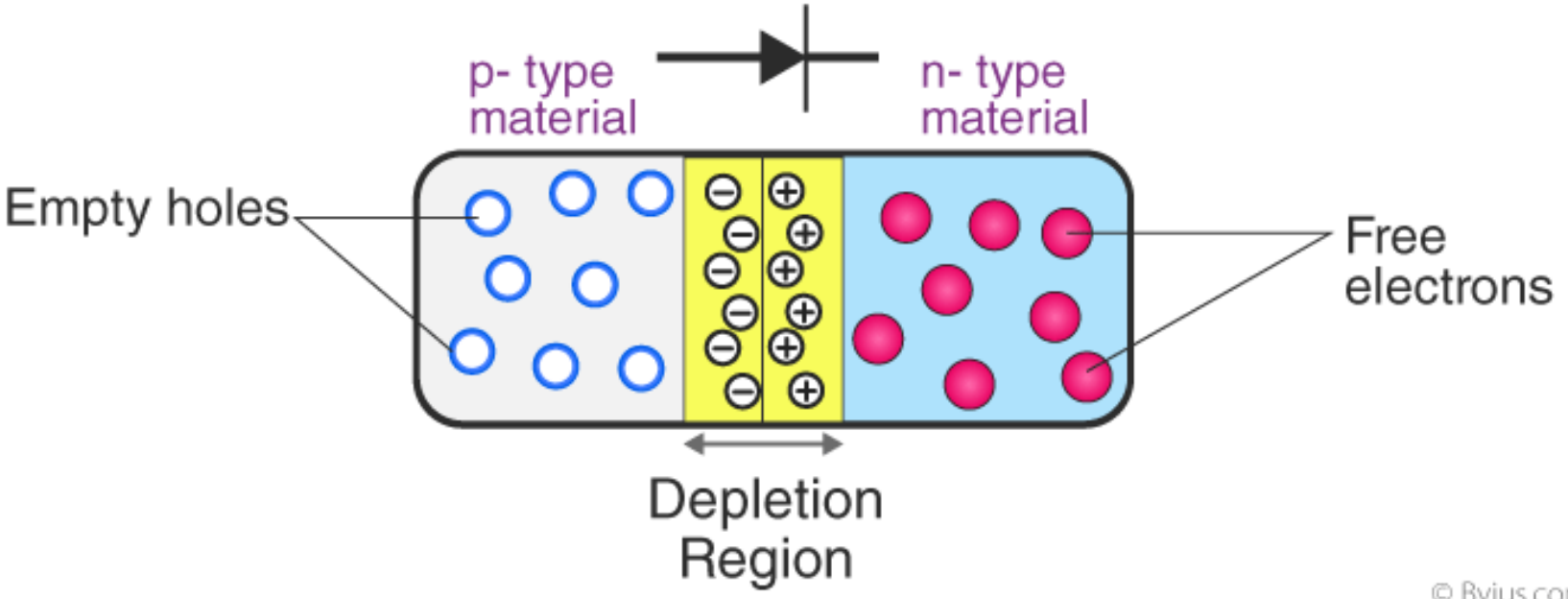


What is a Diode?



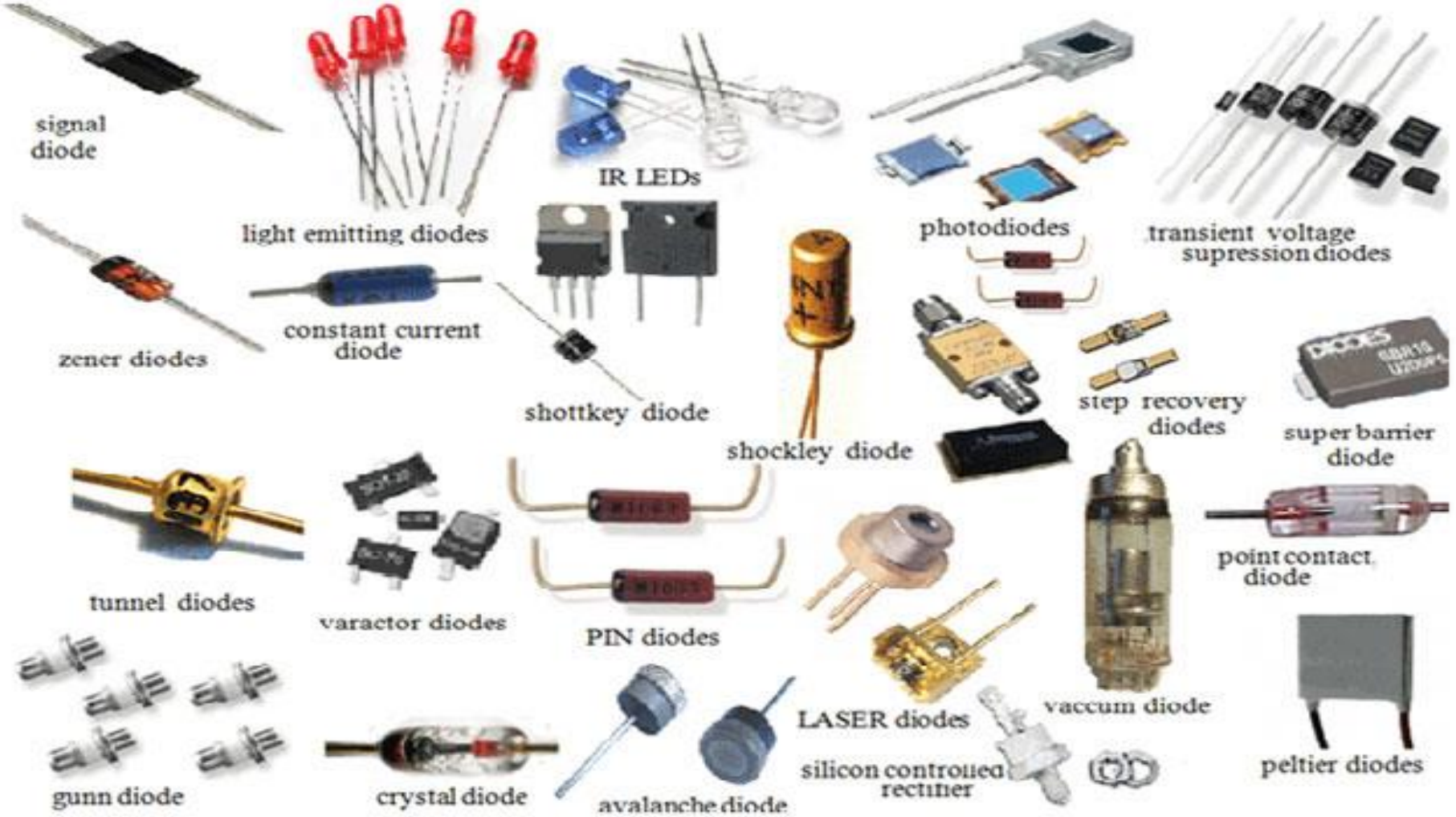
What is P-N Junction?

A P-N junction is an interface or a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.

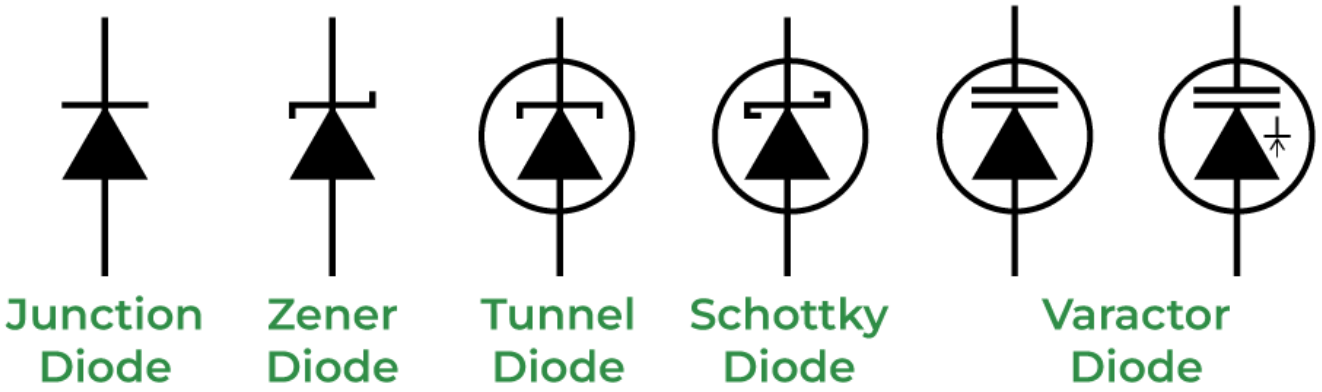


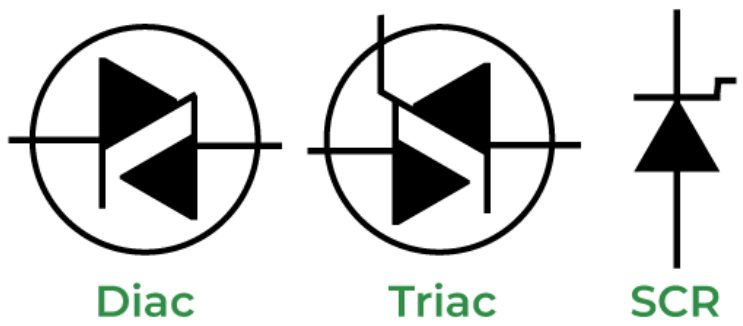
© Byjus.com

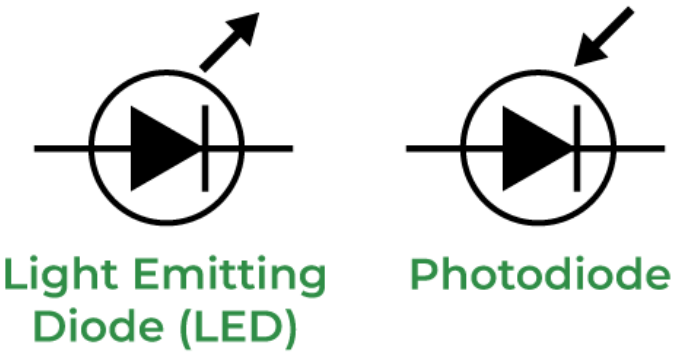
Types of Diodes



Types of Diode





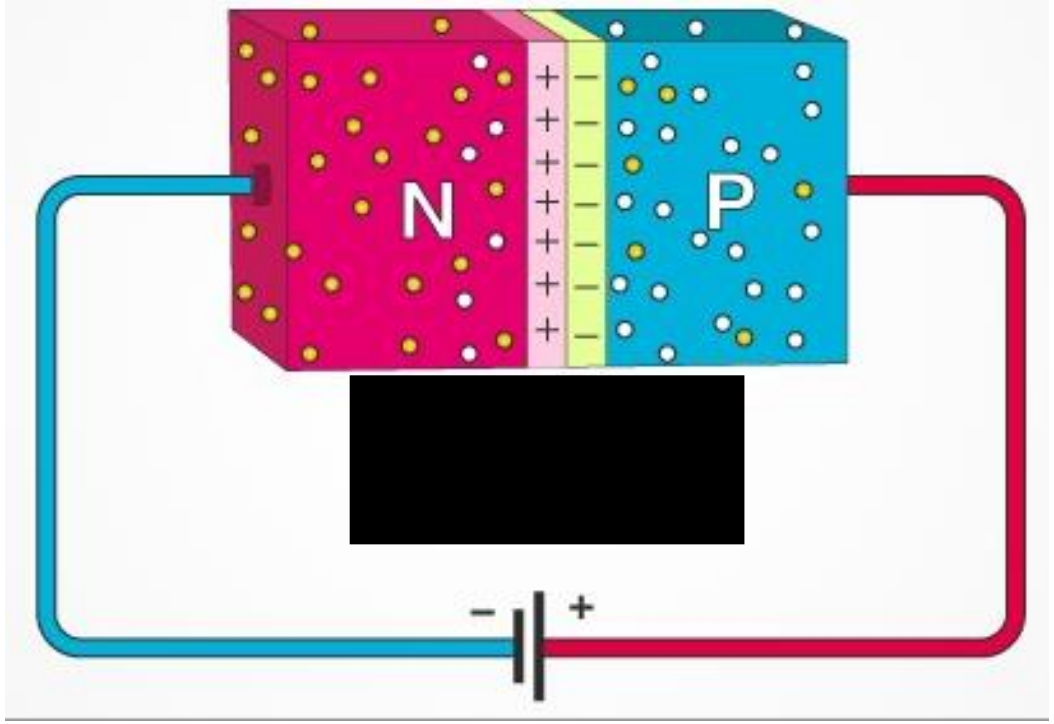


Rectifier Diode

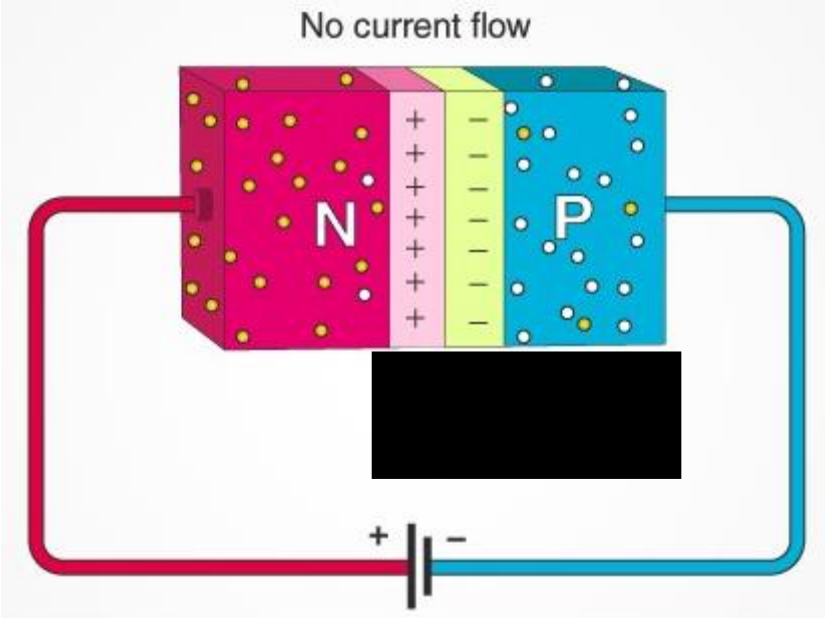


Diode modes

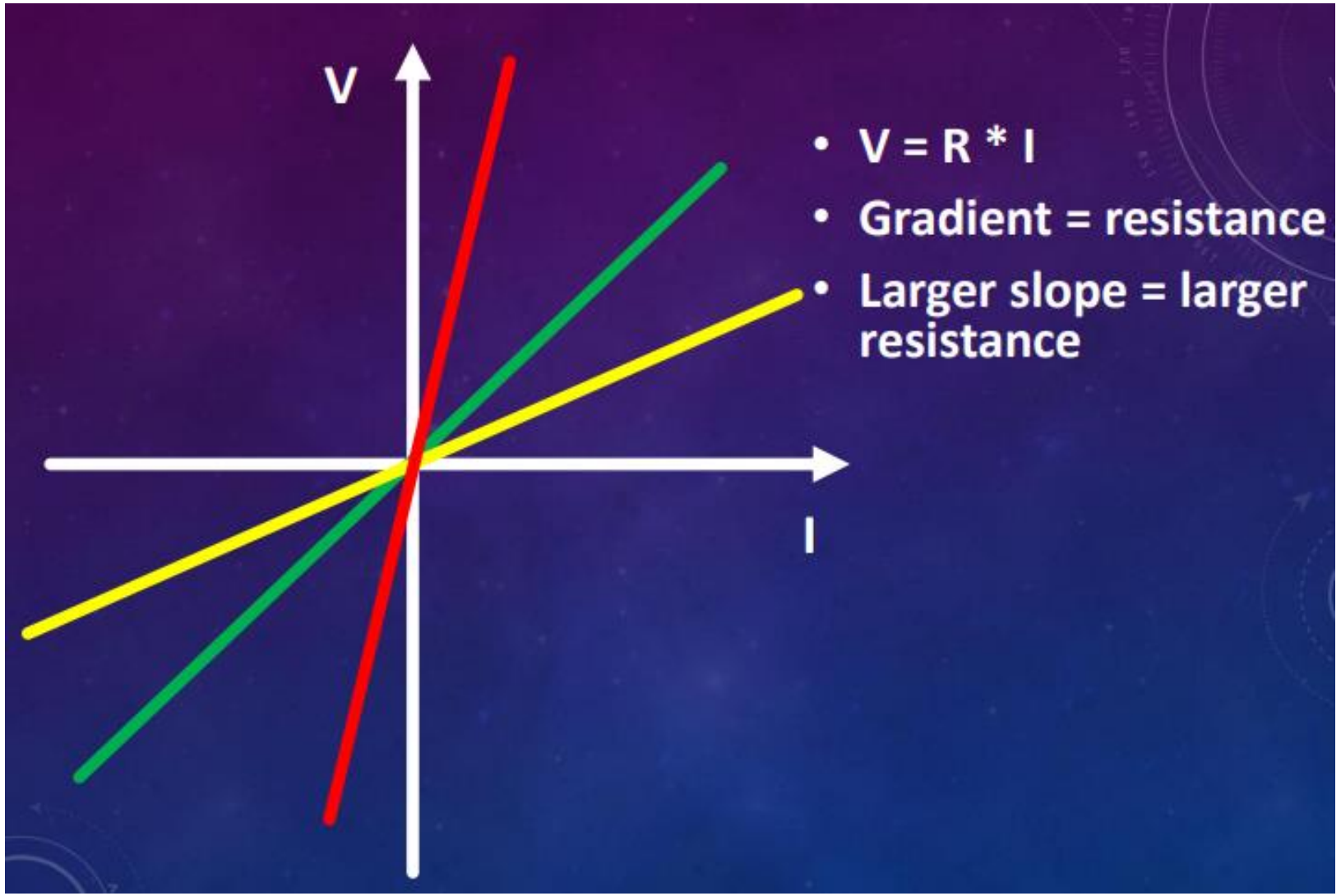
1. **Forward Bias**



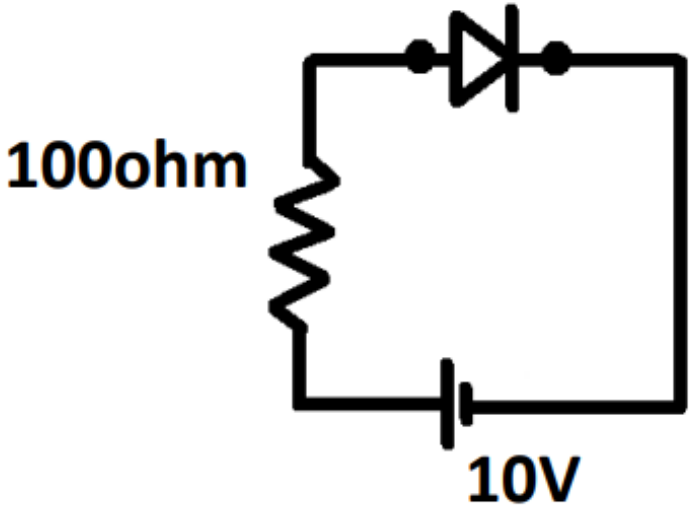
2. Reverse Bias



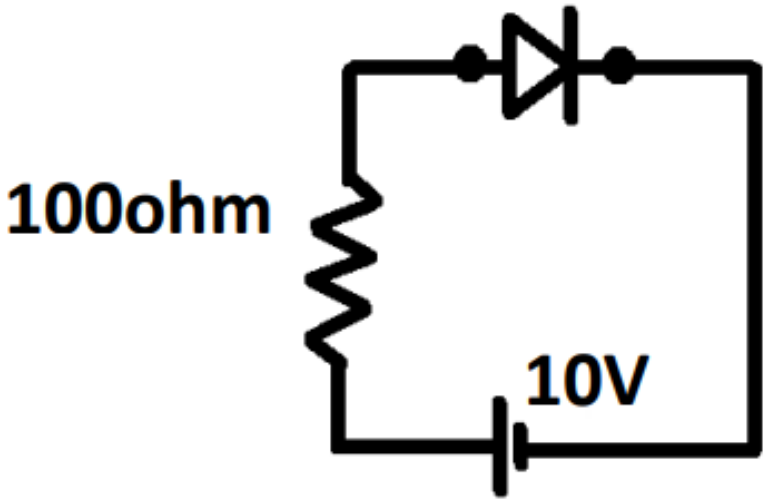
Resistor I-V Characteristics



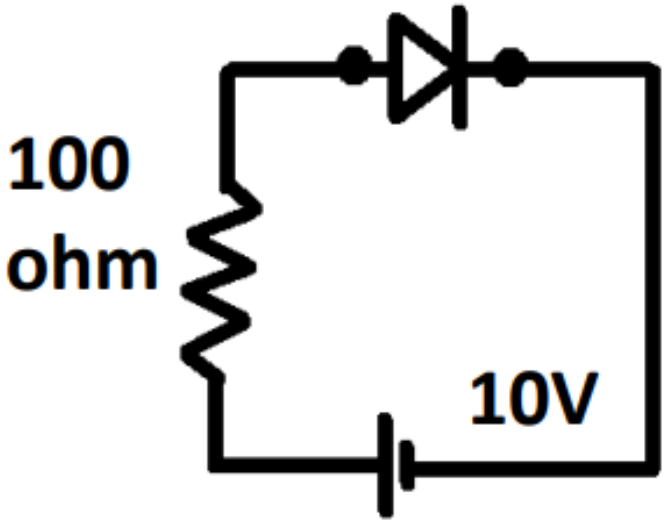
Q3) Considering the diode is a ideal one, calculate the current through the diode. Then draw an equivalent circuit and graph.



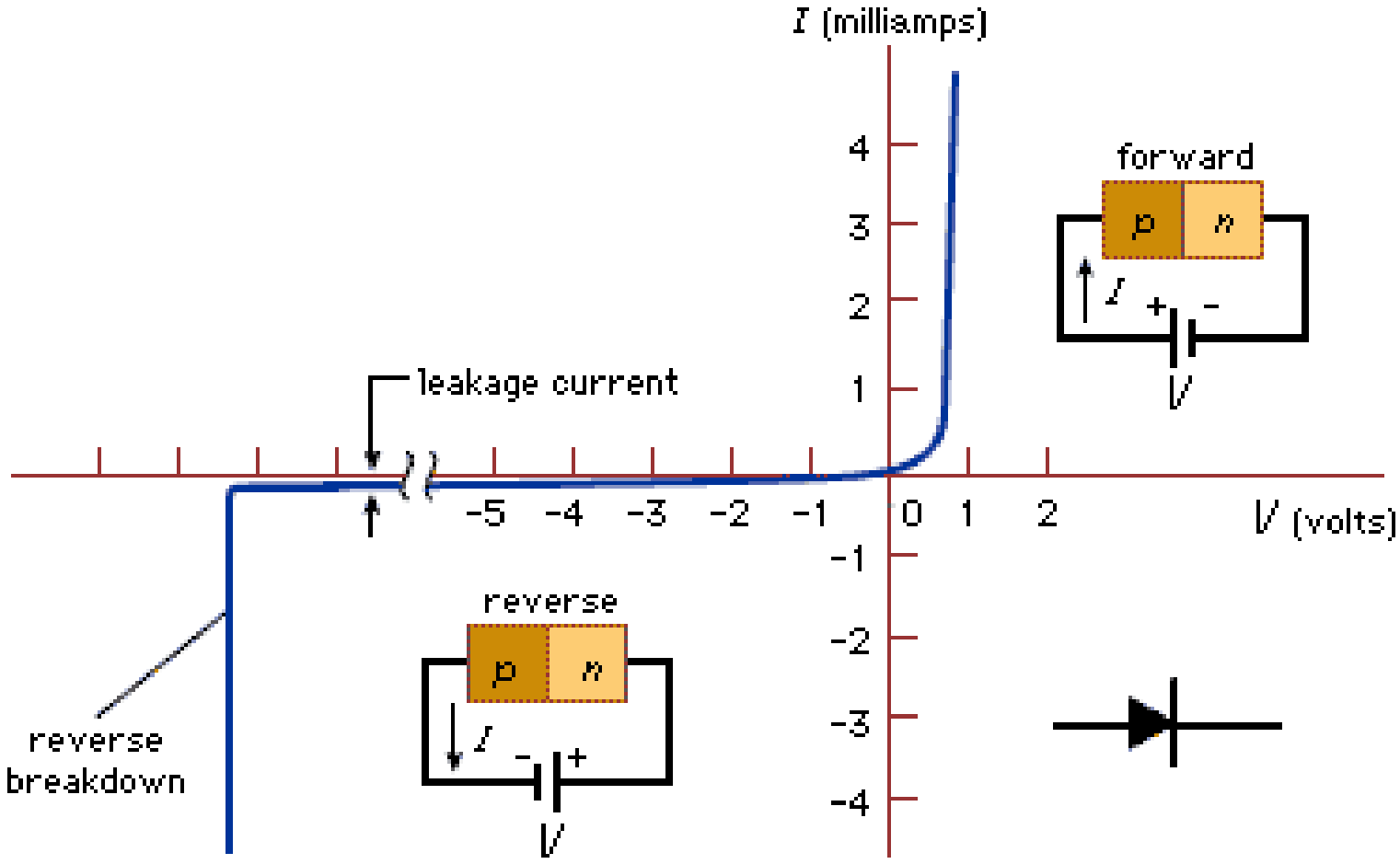
Q4) Calculate the current through the Si diode. Then draw an equivalent circuit and graph. (0.7 V Silicon)



Q5) Calculate the current through the Si diode when the diode has a resistance of 25 ohm. Then draw an equivalent circuit and graph.



Diode I-V Characteristics



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Applications

Common Diodes
1N4001-7 silicon power diode

Characteristic	Symbol	1N4001	1N4002	1N4003	1N4004	1N4005	1N4006	1N4007	Unit
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	50	100	200	400	600	800	1000	V
RMS Reverse Voltage	$V_{R(RMS)}$	35	70	140	280	420	560	700	V
Average Rectified Output Current (Note 1) @ T_A =+75°C	I_O	1.0							A
Non-Repetitive Peak Forward Surge Current 8.3ms Single Half Sine-Wave Superimposed on Rated Load	I_{FSM}	30							A
Forward Voltage @ I_F = 1.0A	V_{FM}	1.0							V
Peak Reverse Current @T_A = +25°C at Rated DC Blocking Voltage @ T_A = +100°C	I_{RM}	5.0 50							μA
Typical Junction Capacitance (Note 2)	C_j	15				8			pF
Typical Thermal Resistance Junction to Ambient	$R_{\theta JA}$	100							K/W
Maximum DC Blocking Voltage Temperature	T_A	+150							°C
Operating and Storage Temperature Range	T_J, T_{STG}	-65 to +150							°C

Notes:
1. Leads maintained at ambient temperature at a distance of 9.5mm from the case.
2. Measured at 1.0 MHz and applied reverse voltage of 4.0V DC.
3. EU Directive 2002/95/EC (RoHS). All applicable RoHS exemptions applied, see EU Directive 2002/95/EC Annex Notes.

Common Diodes

- 1N4001-7 silicon power diode
- 1N34, 1N34A Detector Diodes
- 1N4448 silicon signal diode
- MPN3404 (or BA479) PIN diode
- KV1360NT variable capacitance diode (varicap)

Replacement and Equivalent

You can replace the 1N4004 with the

RL207, RL206, RL205, RL204, HER208, HER207, HER206, HER205, HER158, HER157, HER156, HER155, HER108, HER107, HER106, HER105, FR207, FR206, FR205, FR204, FR157, FR156, FR155, FR154, FR107, FR106, FR105, FR104, EM520, EM516, EM513, BY133, 1U8, 1U7, 1U6, 1U5, 1N5399, 1N5398, 1N5397, 1N5395, 1N4937, 1N4936, 1N4007, 1N4006, 1N4005, 1A7, 1A6, 1A5, 1A4, SF16, SF18, SF26, SF28, UF4004, UF4005, UF4006, UF4007, 1N4004G, 1N4005G, 1N4006G, 1N4007G, 1N4936G, 1N4937G